

Slice & Dice

Pizza Sales Data Analysis & Visualization

PROJECT DOCUMENTATION

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Table of Contents

1	Introduction & Business Context
2	Problem Statement
3	Project Goals
4	Data Overview & Tech Stack
5	Data Preprocessing & Cleaning using SQL
6	Analysis Results, Visualization and Dashboard
7	Exploratory Data Analysis (EDA) using SQL
8	Exploratory Data Analysis and Business Insights
9	Final Recommendations & Conclusion

1.Introduction

This project dives deeper into a pizzeria's sales data and helps us understand what makes the business function successfully. Every time a customer purchases a pizza, a lot of details are recorded in the transaction like the pizza name, size, category and its cost.

By analyzing the pizza sales information, we can spot the patterns and trends that help us understand the business better and by analyzing and understanding the dataset we can find ways to make the pizza sales business more successful.

Business Context

Running a pizza business isn't just about cooking great food but it is also about making smart decisions for the business to run successfully. Particularly good business decisions must be made after analyzing the data regarding the following:

- Which category of pizza brings in more money? This information also helps us to make key decisions for buying items as a part of inventory management.
- Which category of pizza is not very popular? This information will help us to decide whether to continue cooking this pizza in future.
- On which days and hours are the pizzeria busiest? This information helps the managers to allocate the right number of staff.

2. Problem Statement

The pizza restaurant is currently operating on guesswork. They just know that customers come, order food and make the payment. They know that some categories of food are hot favorites while others are not. But they don't know the details like how much food is wasted and which is the most unpopular pizza and which are the peak hours for customer visits. We need accurate and complete information with context, to better run the pizza restaurant business.

To maximize the revenue and customer satisfaction, we should make use of the pizza sales dataset for analysis and decide upon the following three critical areas of management in the pizza sales business:

- **SALES:** Which pizzas drive revenue, and which are underperforming?
- **STAFFING:** When are the peak hours and peak days that require more kitchen staff?
- **INVENTORY MANAGEMENT:** What are the most popular pizzas and pizza sizes? This information will help us optimize supply ordering.

3. Project Goals

In this pizza sales data analysis project, our main goals will be to gain insights into the following and to produce good business decisions:

- Which pizzas are the top sellers, and which ones should we consider removing from the menu?
- What are the busiest days and hours of the week?
- What size and category of pizza do customers prefer the most?
- What are all the categories of pizzas from the Pizza Sales Database?
- What are all the categories of pizzas and list out the pizza names under each category?
- What are all the pizza sizes available?
- Give out all the distinct pizza names and the many sizes in which they are available?
- What is the first order date and the last order date? What is the total number of days between these orders?
- What is the Total Revenue?
- What is the total number of pizzas sold?

- What is the average price of pizzas?
- What is the total number of orders?
- What is the total number of pizzas?
- What are the key metrics of business?
- What are the total orders by pizza category?
- What is the total number of pizzas by pizza category?
- What is the average cost by pizza category?
- What is the total revenue generated by each category?
- What is the total quantity of pizzas sold by each category?
- Which five pizzas generate the highest revenue?
- What are the five worst performing pizzas in terms of sales?
- Display the pizza ranking.
- Estimate the pizza demand based on immediate past trend.

4.Data Overview & Tech Stack

1. Dataset Summary

The dataset was downloaded from Kaggle. It is a pizza sales dataset. It tracks the data for individual pizzas that were sold.

Total Number of Records: 2036

Total Number of Columns: 12

Datatypes Included: INTEGER, DECIMAL, VARCHAR, DATE, TIME

2. Data Structure & Columns

The columns in the dataset are 12 in number and can be divided into four logical categories and they are as follows:

a. Order and Transaction Identifiers

Fields	Data Type	Description
pizza_id	int	A unique ID for each individual pizza row in the dataset.
order_id	int	A unique ID linking multiple pizzas bought together in a single checkout transaction.

b. Product and Menu Information

Fields	Data Type	Description
pizza_name_id	varchar()	The identifier for the specific pizza type.
pizza_name	varchar()	The official, customer-facing name of the pizza.
pizza_category	enum()	The menu grouping.
pizza_size	char()	The size of the pizza offered.
pizza_ingredients	varchar()	A comma-separated list of the toppings used on that specific pizza.

c. Time and Date Fields

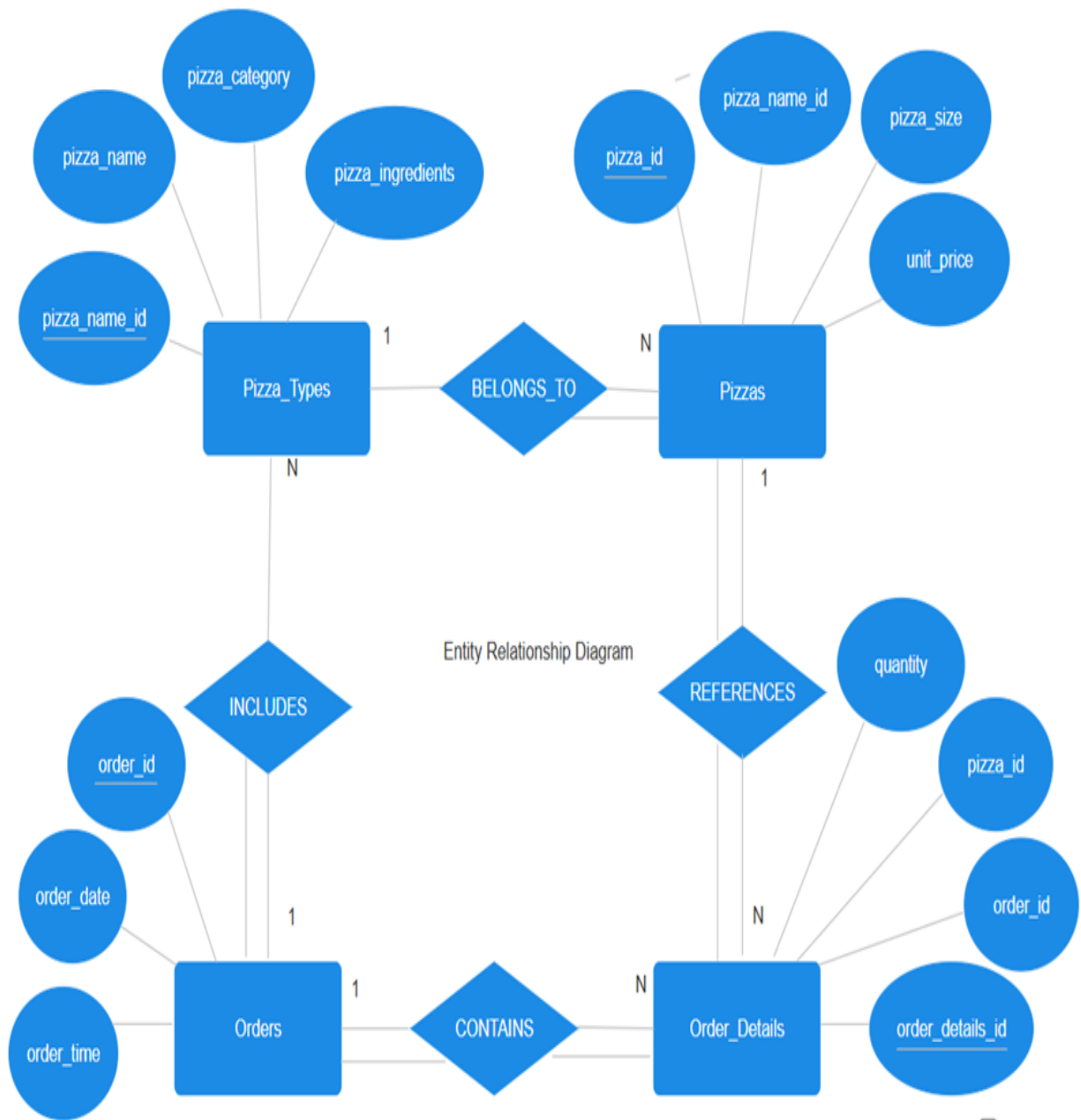
Fields	Data Type	Description
order_date	date	The day the purchase was made (YYYY-MM-DD).
order_time	time	The exact time, the order was placed (HH:MM:SS).

d. Financials and Quantities

Fields	Data Type	Description
quantity	int	The number of items of this exact pizza size/type ordered in this specific line.
unit_price	decimal()	The cost of a single pizza.
total_price	decimal()	Calculated per row as: quantity * unit_price

3. Database Schema

Conceptual Database Schema



In the above ER diagram, there are four entities which are represented each by a rectangle. These four entities are the four database tables. The oval shape symbols connected to each entity are their respective attributes. The underlined attributes represent the primary keys of the respective database tables.

Types of Relationship between the Tables:

- **Pizzas to Pizza_Types:** They share a **many-to-one** relationship. Many pizzas can belong to a single category like Chicken.
- **Order_Details to Pizzas:** They share a **many-to-one** relationship. Many Order_Details can reference the same pizza_name_id like Hawaiian_m.
- **Orders to Order_Details:** They share a **one-to-many** relationship. A single order can be placed for pizzas of different sizes and numbers. For example, I can place a single order of three pizzas wherein two pizzas are of size L and one pizza is of size S.
- **Orders to Pizza_Types:** They share a **one-to-many** relationship. Many pizzas of different categories and of different ingredients can be placed in a single, unique order.

4. Logical Database Schema

In this section, we describe the various tables with their table names, attributes and datatypes. There are about 4 tables with the names: Pizza_Types, Pizzas, Orders and Order_Details. The prefix **PK** in front of an attribute signifies that the attribute is a primary key. And the prefix **FK** in front of an attribute signifies that the attribute is a foreign key.

Pizza_Types

Attribute	Datatype
PK pizza_name_id	varchar(45)
pizza_name	varchar(255)
pizza_category	enum()
pizza_ingredients	varchar(500)

Pizzas

Attribute	Datatype
PK pizza_id	int
FK pizza_name_id	varchar(45)
pizza_size	char(1)
unit_price	decimal(10,2)

Orders

Attribute	Datatype
PK order_id	int
order_date	date
order_time	time

Order_Details

Attribute	Datatype
PK order_details_id	int
FK order_id	int
FK pizza_id	int
quantity	int

5. Software Specifications

Tools Used:

- MySQL Server
- MySQL Workbench
- Power BI Desktop

Programming Language Used:

- SQL

6. Implementation 1: Data Cleaning

- a. Downloaded the dataset from Kaggle.
- b. Completed **data cleaning** on the dataset using MySQL.
- c. Ran SQL queries on the dataset for analyzing the Pizza Sales Business.
- d. Finally visualized the data and developed a dashboard using Power BI.

7. Implementation 2: Exploratory Data Analysis

- a) **Normalized** the Cleaned Pizza Sales Analysis Dataset into many tables.
- b) Performed **Exploratory Data Analysis** on the same using MySQL.

5.Data Preprocessing & Cleaning using SQL

Below are the data cleaning steps that I followed to produce clean analysis regarding Pizza Sales.

STEPS:

1. Imported the Pizza Sales Dataset into MySQL Workbench.
2. Care was taken to import order_date column as VARCHAR() and was later changed to DATE data type. This method was followed to successfully import order_date column without issues.
3. Cleaned the order_date column. Used IS NOT NULL, IS NULL, TRIM(order_date) to fix the data. Formatted the order_date column values to YYYY_MM_DD format.
4. Modified unit_price column and total_price column to DECIMAL(10,2).
5. Replaced \$ signs/symbols in unit_price and total_price columns with a space.
6. Performed Math Sanity Check on the column total_price to ensure that $\text{quantity} * \text{unit_price} = \text{total_price}$.
7. Modified the order_time column to TIME datatype.
8. Checked for anomalies in the order timings. This ensured that all orders were placed between 11 AM to 10 PM during the Pizzeria's working hours.
9. Checked for duplicate values in the pizza_id column. This ensures that your total pizza count won't be artificially inflated.
10. Standardized the pizza_name_id column by updating `pizza_name_id = TRIM(LOWER(pizza_name_id))`; The names in this column ended with _m and _l wherein _m signifies that the pizza is of medium size and _l signifies that the pizza is of large size. Any space like Hawaiian m, will suggest that hawaiian m is a different pizza type. To ensure correctness we use this step.

11. The `pizza_ingredients` column contains comma separated values. Hidden spacing in this column was cleaned.
12. Performed a NULL AUDIT check on the columns `pizza_id`, `order_id`, `quantity` and `total_price`. Presence of NULL values will make the data dirty, and it will skew our analysis. This NULL AUDIT check made sure that there were no NULL values in the columns.
13. Standardized categorical fields like `pizza_size`, `pizza_category` and `pizza_name` that are present in the dataset. It is important to standardize the values of these columns in order to come up with accurate analysis. Standardized the sizes. Made sure only M, L and S pizza sizes are present because sometimes text imports mess up with casing and spaces. Hence, we need to standardize them. Also ensured that the `pizza_category` column values are uniformly capitalized.

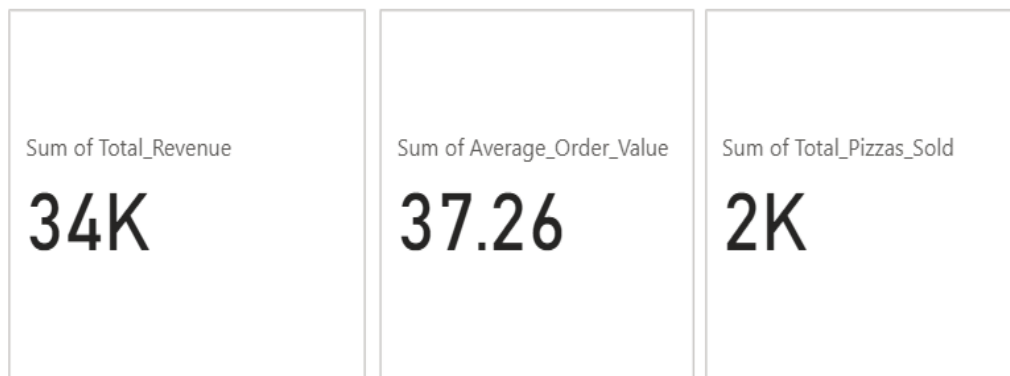
6. Analysis Results, Visualization and Dashboard

Some important analysis that I came up with after data cleaning are as follows:

1. Sales Performance Analysis

The following KPIs or Key Performance Indicators were tracked:

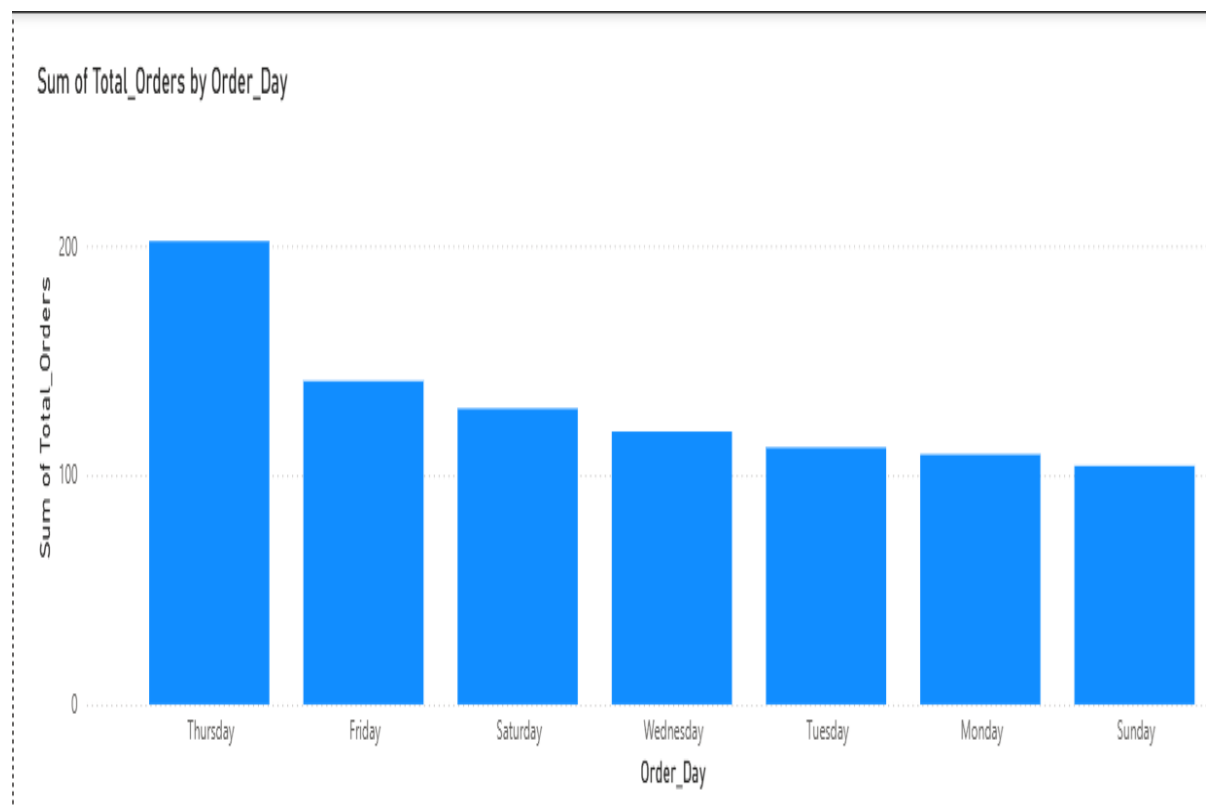
- **Total Revenue:** The Total Revenue generated was 34.13K dollars.
- **Average Order Value:** It tells how much customers spend per order on average? It was found that customers spent an average of 37.26\$ per order. It is the Average Order Value.
- **Total Pizzas Sold:** It is the sum of the quantities. It was found that nearly a total of 2K pizzas were sold.



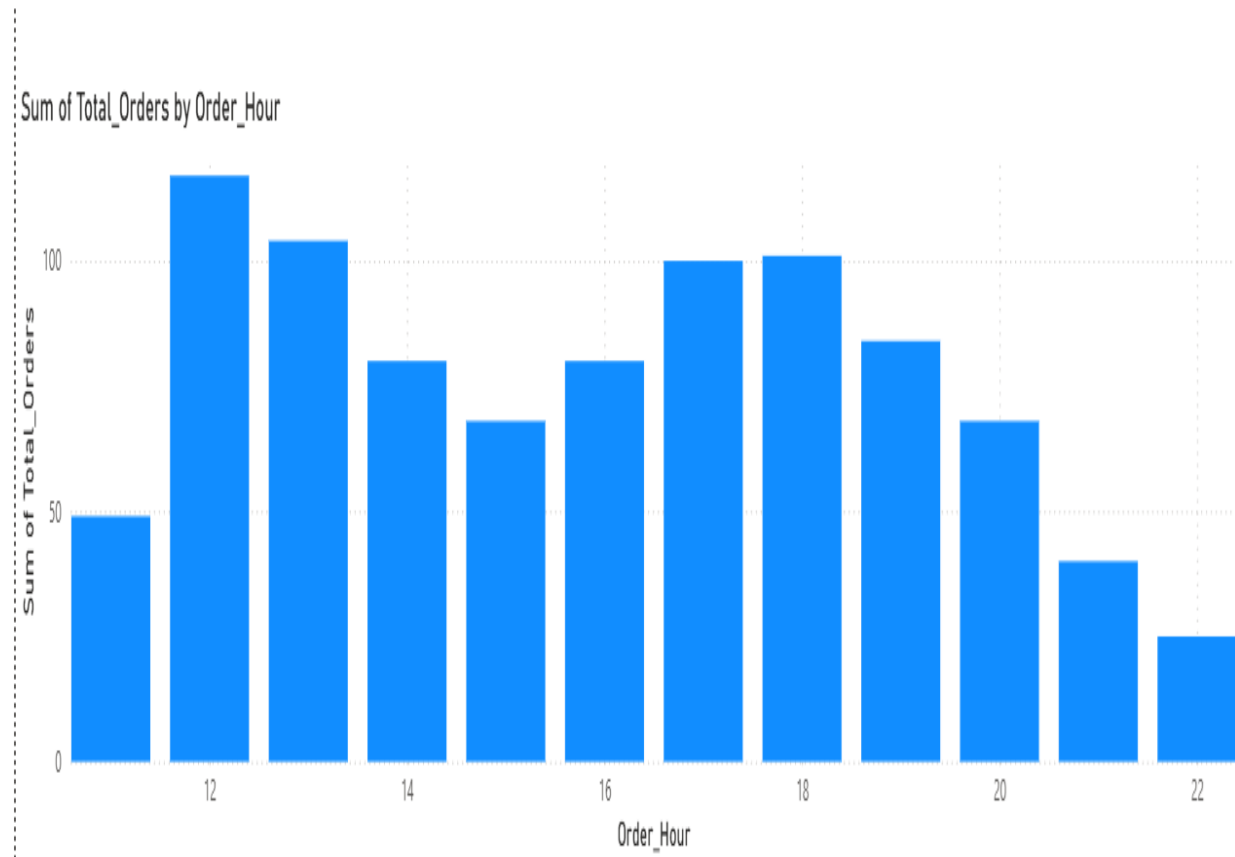
2. Time-Based Analysis

With Time-Based Analysis, we tracked the following information:

- **Daily Trend for Total Orders:** From the chart below, we notice that Thursdays were the favorite days for ordering pizzas.



- **Hourly Trend for Total Orders:** From the chart below, we notice that the peak pizza ordering hours are at 12 noon and at 6 PM in the evening.

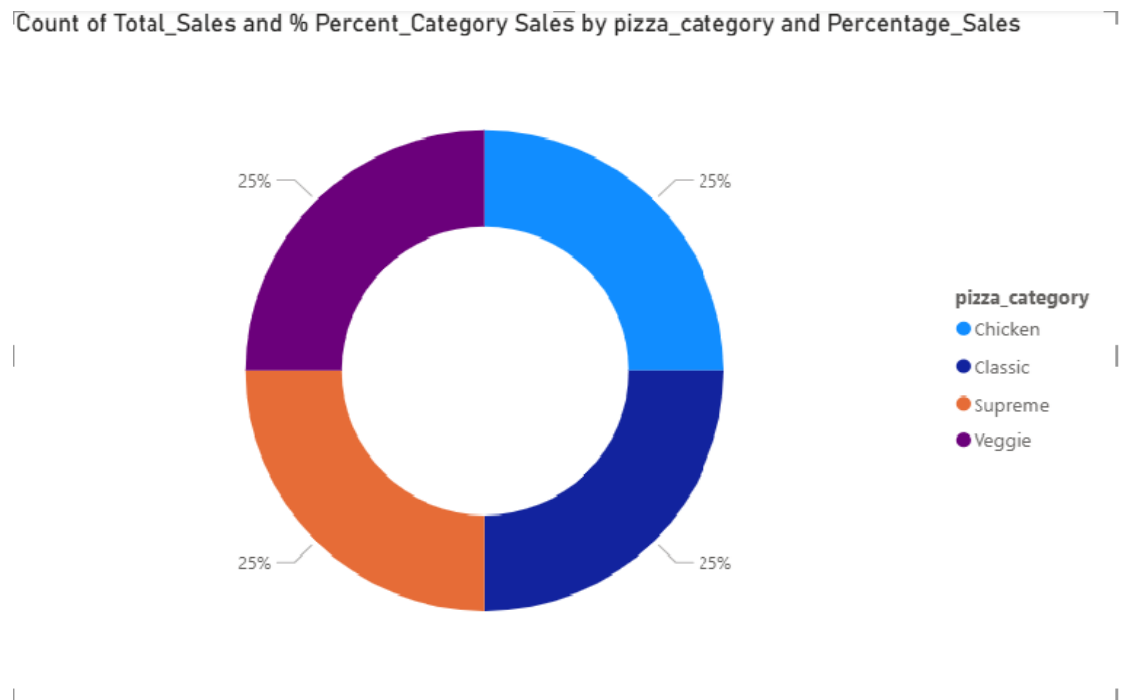


3. Product and Category Analysis

The donut and pie charts below help us understand how the business runs based on various dimensions like Pizza Category and Pizza Size. The information below can help us make good business decisions.

- **Percentage of sales by Pizza Category**

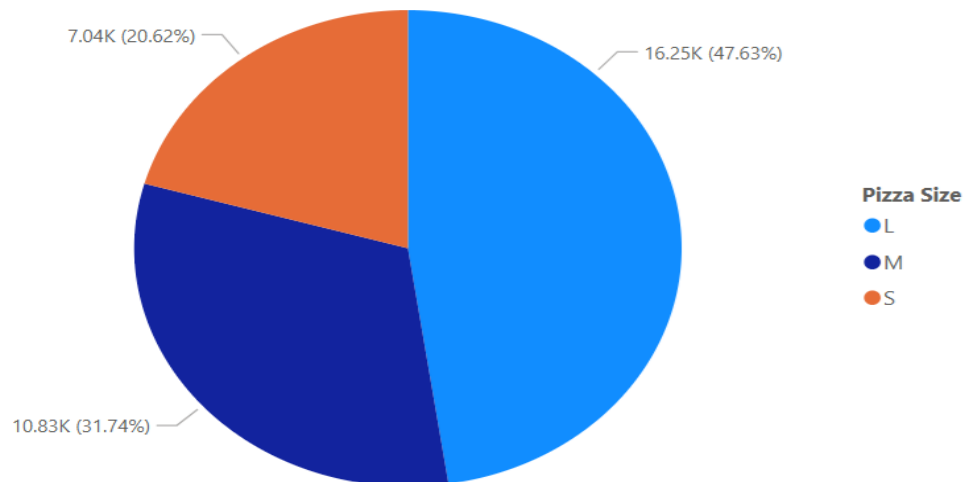
From the donut chart below, it can be observed that each Pizza Category contributes to nearly 25 % of Pizza Sales.



- **Total Sales by Pizza Size**

From the pie chart below, it can be observed that pizza size L (Large) contributes to about 47.63 % of sales, which is the largest. Followed by pizza size M (Medium), which contributes to about 31.74% of sales and lastly the pizza size S, which is small, contributes to about 20.62% of pizza sales.

Total_Sales by Pizza Size



- **Top 5 best sellers by revenue**

The table below shows the Top 5 Pizzas that contributed maximum to the Pizzeria's Revenue.

pizza_name_id	Total_Revenue
five_cheese_I	1,332.00
thai_ckn_I	1,245.00
spicy_ital_I	1,099.75
bbq_ckn_I	933.75
southw_ckn_I	913.00

Based on the clean pizza sales dataset and analysis results above, a **Pizza Sales Analysis Dashboard** has been created using Power BI. The dashboard has 4 pages. First page is dedicated to Pizza Sales Analysis. Second page is dedicated to Sales Performance Analysis. Third page is dedicated to Time-Based Analysis and the fourth page is dedicated to Product and Category Analysis.

7.Exploratory Data Analysis (EDA) using SQL

To further understand the Pizza Sales Business better, we dive deeper into analyzing by applying **Exploratory Data Analysis** to the Pizza Sales Database using SQL programming language.

To continue with Exploratory Data Analysis, the Pizza Sales dataset was first cleaned and secondly normalized into many tables. This is again a pure SQL-Driven Exploratory Analysis.

There are four normalized tables used for this analysis. Below are the 4 tables with their attribute names and datatypes.

Pizza_Types

Attribute	Datatype
PK pizza_name_id	varchar(45)
pizza_name	varchar(255)
pizza_category	enum()
pizza_ingredients	varchar(500)

Pizzas

Attribute	Datatype
PK pizza_id	int
FK pizza_name_id	varchar(45)
pizza_size	char(1)
unit_price	decimal(10,2)

Orders

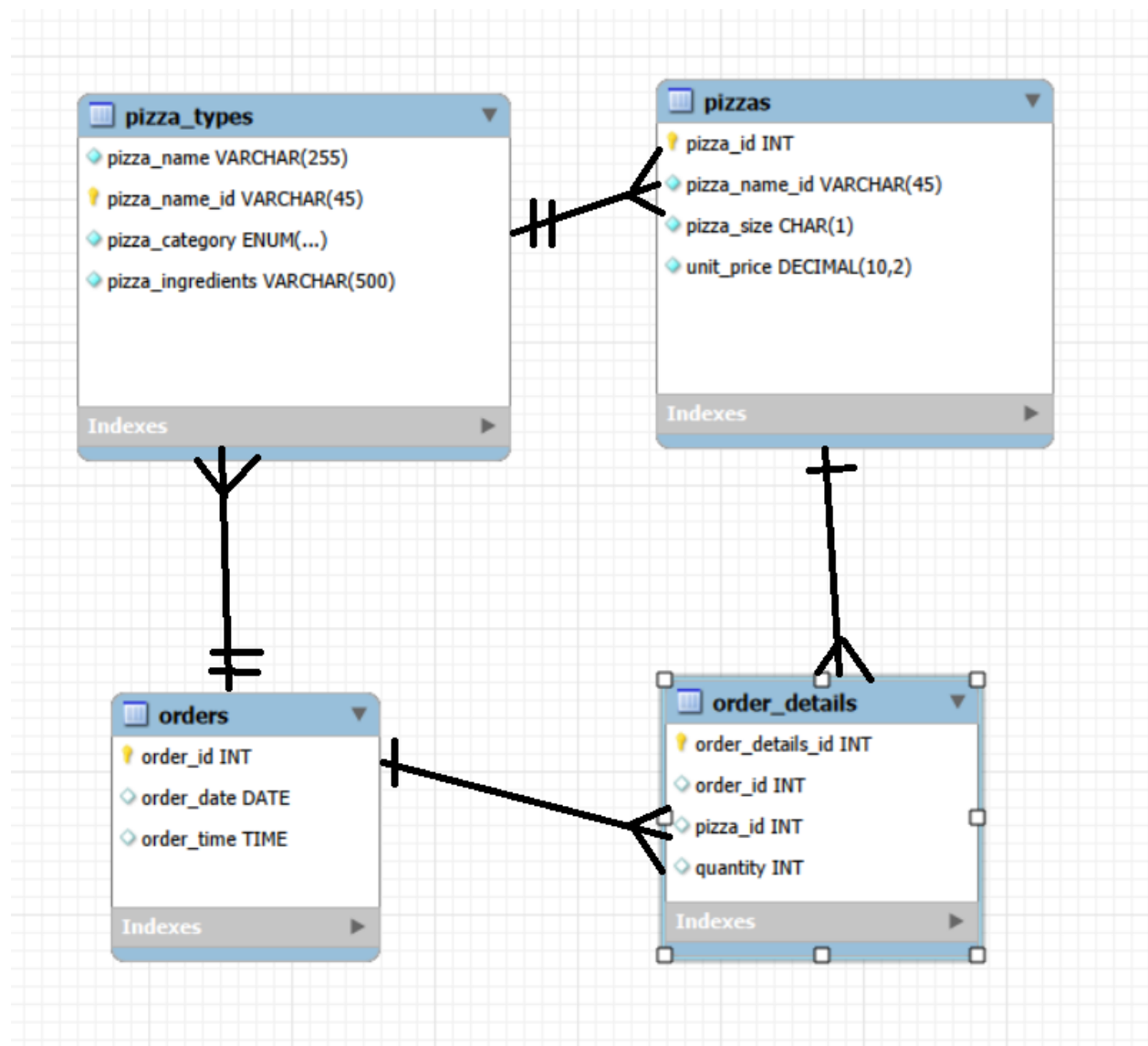
Attribute	Datatype
PK order_id	int
order_date	date
order_time	time

Order_Details

Attribute	Datatype
PK order_details_id	int
FK order_id	int
FK pizza_id	int
quantity	int

Entity-Relationship Diagram

Below is the Entity- Relationship Diagram for the normalized Pizza Sales Database. It is known as Crow's Foot Notation.

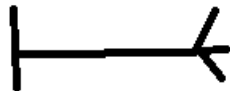


Crow's Foot Notation Explanation:

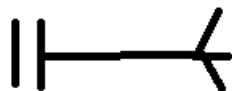
It is essentially used to explain the Entity-Relationship Diagram, wherein the diagram maps out the relationship between various tables/entities in a database showing the cardinality between the relationships. Below are the cardinality relationships explained along with their symbols.



EXACTLY ONE



ONE TO MANY



EXACTLY ONE TO MANY

In the above ER-Diagram, you can notice the above symbols drawn between tables, which tell us that the tables are connected. Each rectangular box represents an entity. The entities/tables in the diagram are as follows:

- **pizza_types**
- **pizzas**
- **orders**
- **order_details**

The other elements present inside the rectangular box are the attributes of the respective tables.

Data Cleaning and Preparation

- ❖ The Pizza Sales Dataset used for Exploratory Data Analysis is a clean dataset.
- ❖ NULL values were removed.
- ❖ The `order_date` and `order_time` column were converted to the DATE and TIME formats, respectively.
- ❖ Duplicates were handled in the dataset.

8. Exploratory Data Analysis and Business Insights

Dimensions Exploration

This section deals with exploring the various dimensions in the Pizza Sales Database.

1. What are all the categories of pizzas from the Pizza Sales Database?

SQL Query

```
SELECT DISTINCT(pizza_category)
FROM pizza_types
```

Result:

pizza_category
Chicken
Classic
Supreme
Veggie

Insight: From the above table, we notice that there are four pizza categories. And they are Chicken, Classic, Supreme and Veggie.

2. What are all the categories of pizzas and also list out the pizza names under each category?

SQL Query

```
SELECT DISTINCT pizza_category, pizza_name  
FROM pizza_types
```

Result:

pizza_category	pizza_name
Chicken	The Barbecue Chicken Pizza
Classic	The Big Meat Pizza
Supreme	The Brie Carre Pizza
Supreme	The Calabrese Pizza
Chicken	The California Chicken Pizza
Chicken	The Chicken Alfredo Pizza
Chicken	The Chicken Pesto Pizza
Classic	The Classic Deluxe Pizza
Veggie	The Five Cheese Pizza
Veggie	The Four Cheese Pizza
Veggie	The Green Garden Pizza
Classic	The Hawaiian Pizza
Classic	The Italian Capocollo Pizza
Supreme	The Italian Supreme Pizza
Veggie	The Italian Vegetables Pizza
Veggie	The Mediterranean Pizza
Veggie	The Mexicana Pizza
Classic	The Napolitana Pizza
Classic	The Pepperoni, Mushroom, and Peppers Pizza
Classic	The Pepperoni Pizza
Supreme	The Pepper Salami Pizza
Supreme	The Prosciutto and Arugula Pizza
Supreme	The Sicilian Pizza
Supreme	The Soppressata Pizza
Chicken	The Southwest Chicken Pizza
Supreme	The Spicy Italian Pizza
Veggie	The Spinach Pesto Pizza
Veggie	The Spinach and Feta Pizza
Supreme	The Spinach Supreme Pizza

Chicken
Classic
Veggie

The Thai Chicken Pizza
The Greek Pizza
The Vegetables + Vegetables Pizza

Insight: From the above table, we notice that each of the four pizza categories offer different types of pizzas that are called by different names.

3. What are all the pizza sizes available?

SQL Query

```
SELECT DISTINCT(pizza_size)
FROM pizzas
```

Result:

pizza_size
M
L
S

Insight: The above table tells us that the pizzeria offers pizzas only of three sizes and they are M – Medium, L – Large and S – Small.

4. Give out all the distinct pizza names and the many sizes in which they are available?

SQL Query

```
SELECT DISTINCT pizza_name_id, pizza_size  
FROM pizzas
```

Result:

pizza_name_id	pizza_size
hawaiian_m	M
classic_dlx_m	M
five_cheese_l	L
ital_supr_l	L
mexicana_m	M
thai_ckn_l	L
ital_supr_m	M
prsc_argla_l	L
bbq_ckn_s	S
the_greek_s	S
spinach_supr_s	S
classic_dlx_s	S
green_garden_s	S
ital_cpcllo_l	L
ital_supr_s	S
mexicana_s	S
spicy_ital_l	L
spin_pesto_l	L
veggie_veg_s	S
mexicana_l	L
southw_ckn_l	L
bbq_ckn_l	L
cali_ckn_l	L
cali_ckn_m	M
pepperoni_l	L
cali_ckn_s	S
ckn_pesto_l	L
big_meat_s	S

soppressata_l	L
four_cheese_l	L
napolitana_s	S
calabrese_m	M
four_cheese_m	M
ital_veggie_s	S
mediterraneo_m	M
peppr_salami_s	S
spinach_fet_l	L
napolitana_l	L
sicilian_l	L
ital_cpcllo_m	M
southw_ckn_s	S
bbq_ckn_m	M
pepperoni_m	M
prsc_argla_s	S
sicilian_m	M
veggie_veg_l	L
ckn_alfredo_s	S
pepperoni_s	S
green_garden_l	L
green_garden_m	M
pep_msh_pep_l	L
hawaiian_s	S
peppr_salami_m	M
ckn_alfredo_m	M
peppr_salami_l	L
spin_pesto_s	S
thai_ckn_m	M
classic_dlx_l	L
ckn_pesto_m	M
hawaiian_l	L
pep_msh_pep_s	S
spinach_supr_m	M
prsc_argla_m	M
mediterraneo_l	L
southw_ckn_m	M
pep_msh_pep_m	M
sicilian_s	S
spicy_ital_s	S
thai_ckn_s	S
spinach_supr_l	L
ital_veggie_l	L
veggie_veg_m	M

the_greek_m	M
ckn_pesto_s	S
spinach_fet_s	S
spicy_ital_m	M
ital_veggie_m	M
ital_cpcllo_s	S
mediterraneo_s	S
spinach_fet_m	M
napolitana_m	M
spin_pesto_m	M
brie_carre_s	S
ckn_alfredo_l	L
calabrese_s	S
the_greek_l	L
soppressata_m	M
soppressata_s	S
calabrese_l	L

Insight: The above table lists all the unique pizzas available in the pizzeria with their respective sizes, each of M, S and L.

Date Exploration

This section deals with date exploration.

1. What is the first order date and the last order date? What are the total number of days between these orders?

SQL Query

```
SELECT MIN(order_date) as first_order_date,
       MAX(order_date) as last_order_date,
       DATEDIFF(MAX(order_date),MIN(order_date))as
Total_No_of_Days_between_Orders
```

FROM orders

Result:

first_order_date	last_order_date	Total_No_of_Days_between_Orders
1/1/2015	1/16/2015	15

Insight: The above result conveys when the first pizza order was placed and also conveys when the last pizza order was placed. It also gives out the total number of days that passed between these pizza orders.

Measures Exploration

This section deals with calculating the key metrics of the business by applying aggregate functions.

1. What is the Total Revenue?

SQL Query

```
SELECT SUM(od.quantity * p.unit_price) as Total_Revenue
FROM pizzas as p inner join order_details as od
on p.pizza_id=od.pizza_id
```

Result:

Total_Revenue
34126

Insight: The above result gives out the total revenue earned by the pizzeria.

2. What are the total number of pizzas sold?

SQL Query

```
SELECT SUM(quantity) as Total_Quantity  
FROM order_details
```

Result:

Total_Quantity
2077

Insight: 2,077 were the total number of pizzas sold by the Pizzeria.

3. What is the average price of pizzas?

SQL Query

```
SELECT AVG(unit_price) as Average_Price  
FROM pizzas
```

Result:

Average_Price
16.422692

Insight: The above result tells us that the average price of each pizza that was sold is nearly 16\$.

4. What are the total number of orders?

SQL Query

```
SELECT COUNT(DISTINCT order_id) as Total_Orders  
FROM orders
```

Result:

Total_Orders
916

Insight: The total number of orders that were placed for pizzas in the pizzeria was 916.

5. What is the total number of pizzas?

SQL Query

```
SELECT COUNT(DISTINCT pizza_id) as Total_No_of_Pizzas  
FROM pizzas
```

Result:

Total_No_of_Pizzas
2036

Insight: The total number of available pizzas in the pizzeria is 2,036.

Report on Key Metrics of the Business

Below is the SQL Code used for generating the report to show the Key Metrics of the Business:

```
SELECT 'Total Sales' as measure_name, SUM(od.quantity * p.unit_price) as measure_value
FROM pizzas as p inner join order_details as od
on p.pizza_id=od.pizza_id
UNION ALL
SELECT 'Total Quantity' as measure_name, SUM(quantity) as measure_value
FROM order_details
UNION ALL
SELECT 'Average Price' as measure_name, AVG(unit_price) as measure_value
FROM pizzas
UNION ALL
SELECT 'Total Orders' as measure_name, COUNT(DISTINCT order_id) as measure_value
FROM orders
UNION ALL
SELECT 'Total No. of Pizzas' as measure_name, COUNT(DISTINCT pizza_id) as measure_value
FROM pizzas
```

Result:

measure_name	measure_value
Total Sales	34126
Total Quantity	2077
Average Price	16.422692
Total Orders	916
Total No. of Pizzas	2036

Insight: The result we see above is a report generated showing the key metrics of the business. These numbers are some of the most crucial numbers to be monitored in the pizza sales business as it helps us decide on inventory management and waste management by closely monitoring how many pizzas were sold and how many pizzas could not be sold.

Magnitude Analysis for Insights

This section is about comparing the measure values by categories. It helps us understand the importance of various categories.

1. What are the total orders by pizza category?

SQL Query

```
SELECT pizza_category, COUNT(order_id) as Total_Orders  
FROM order_details, pizza_types  
GROUP BY pizza_category  
ORDER BY Total_Orders DESC
```

Result:

pizza_category	Total_Orders
Supreme	50900
Veggie	48864
Classic	44792
Chicken	36648

Insight: The above result gives us the total number of orders per pizza category.

2. What is the total number of pizzas by pizza category?

SQL Query

```
SELECT pizza_category, SUM(pizza_id) as Total_Pizzas
FROM pizza_types as pt inner join pizzas as p
on pt.pizza_name_id = p.pizza_name_id
GROUP BY pizza_category
ORDER BY Total_Pizzas DESC
```

Result:

pizza_category	Total_Pizzas
Classic	581380
Supreme	559607
Veggie	508305
Chicken	447326

Insight: The above result shows us the total number of pizzas present in each category.

3. What is the average cost by pizza category?

SQL Query

```
SELECT pizza_category, AVG(unit_price) as Average_Cost
FROM pizza_types as pt inner join pizzas as p
on pt.pizza_name_id = p.pizza_name_id
GROUP BY pizza_category
ORDER BY Average_Cost DESC
```

Result:

pizza_category	Average_Cost
Chicken	17.825171
Supreme	17.21907
Veggie	16.813185
Classic	14.294627

Insight: The above result shows us the average cost of a pizza per category. Chicken and Supreme category pizzas have average cost higher than Veggie and Classic.

4. What is the total revenue generated by each category?

SQL Query

```
SELECT pt.pizza_category, SUM(od.quantity * p.unit_price) as Total_Revenue
FROM order_details as od inner join pizzas as p
  on od.pizza_id = p.pizza_id
  inner join pizza_types as pt
  on p.pizza_name_id = pt.pizza_name_id
GROUP BY pt.pizza_category
ORDER BY Total_Revenue DESC
```

Result:

pizza_category	Total_Revenue
Supreme	9202.2
Veggie	8437.3
Classic	8414.75
Chicken	8071.75

Insight: The above result summarizes the total revenue earned by each pizza category.

5. What is the total quantity of pizzas sold by each category?

This SQL Query provides us with insights into the distribution of sold pizzas across categories.

SQL Query

```
SELECT pt.pizza_category, SUM(od.quantity) as Total_Quantity_Sold
FROM pizzas as p inner join pizza_types as pt
on p.pizza_name_id=pt.pizza_name_id
inner join order_details as od
on p.pizza_id = od.pizza_id
GROUP BY pt.pizza_category
ORDER BY Total_Quantity_Sold
```

Result:

pizza_category	Total_Quantity_Sold
Chicken	453
Veggie	501
Supreme	534
Classic	589

Insight: From the above result, we realize that Classic category pizza is the largest by quantity sold. Followed by Supreme, Veggie and Chicken.

Ranking Analysis

1. Which 5 pizzas generate the highest revenue?

SQL Query

```
SELECT pt.pizza_name, SUM(od.quantity * p.unit_price) as Total_Revenue
FROM order_details as od inner join pizzas as p
on od.pizza_id = p.pizza_id
inner join pizza_types as pt
on p.pizza_name_id = pt.pizza_name_id
GROUP BY pt.pizza_name
ORDER BY Total_Revenue DESC
LIMIT 5
```

Result:

pizza_name	Total_Revenue
The Thai Chicken Pizza	1808.75
The Barbecue Chicken Pizza	1783
The Italian Supreme Pizza	1684
The Spicy Italian Pizza	1567.25
The Pepperoni Pizza	1548.5

Insight: Out of all the pizza types available, The Thai Chicken Pizza, The Barbecue Chicken Pizza, The Italian Supreme Pizza, The Spicy Italian Pizza and The Pepperoni Pizza generates the highest revenue.

2. What are the 5 worst performing pizzas in terms of sales?

SQL Query

```
SELECT pt.pizza_name, SUM(od.quantity * p.unit_price) as Total_Revenue
FROM order_details as od inner join pizzas as p
on od.pizza_id = p.pizza_id
inner join pizza_types as pt
on p.pizza_name_id = pt.pizza_name_id
GROUP BY pt.pizza_name
ORDER BY Total_Revenue ASC
LIMIT 5
```

Result:

pizza_name	Total_Revenue
The Brie Carre Pizza	307.45
The Mediterranean Pizza	543.75
The Calabrese Pizza	548
The Green Garden Pizza	588.75
The Soppressata Pizza	593.25

Insight: The five worst performing pizzas in terms of sales are The Brie Carre Pizza, The Mediterranean Pizza, The Calabrese Pizza, The Green Garden Pizza and The Soppressata Pizza. These are the slow-moving pizzas and we should consider offering them along with the hot favorites at a discounted price.

3. Display the pizza ranking.

SQL Query

```
WITH RankedPizzas AS
  ( SELECT pt.pizza_name, SUM(od.quantity * p.unit_price) AS Total_Revenue,
    ROW_NUMBER() OVER (ORDER BY SUM(od.quantity * p.unit_price) DESC) AS
rank_pizzas
  FROM order_details AS od INNER JOIN pizzas AS p
    ON od.pizza_id = p.pizza_id
  INNER JOIN pizza_types AS pt
    ON p.pizza_name_id = pt.pizza_name_id
  GROUP BY pt.pizza_name )

SELECT pizza_name, Total_Revenue, rank_pizzas
FROM RankedPizzas
WHERE rank_pizzas <= 5;
```

Result:

pizza_name	Total_Revenue	rank_pizzas
The Thai Chicken Pizza	1808.75	1
The Barbecue Chicken Pizza	1783	2
The Italian Supreme Pizza	1684	3
The Spicy Italian Pizza	1567.25	4
The Pepperoni Pizza	1548.5	5

Insight: The above results show the fast-moving pizzas ranked 1 to 5. These are the favorites and the pizzeria should make sure that they are always in stock.

Forecasting Pizza Demand

The SQL query below calculates the 7 day moving average to project the upcoming Pizza Demand based on immediate past trends.

SQL Query

```
SELECT order_date, pizza_id, quantity,  
AVG(quantity) OVER (PARTITION BY pizza_id ORDER BY order_date ROWS BETWEEN 6  
PRECEDING AND CURRENT ROW) as seven_day_demand_forecast  
FROM sales
```

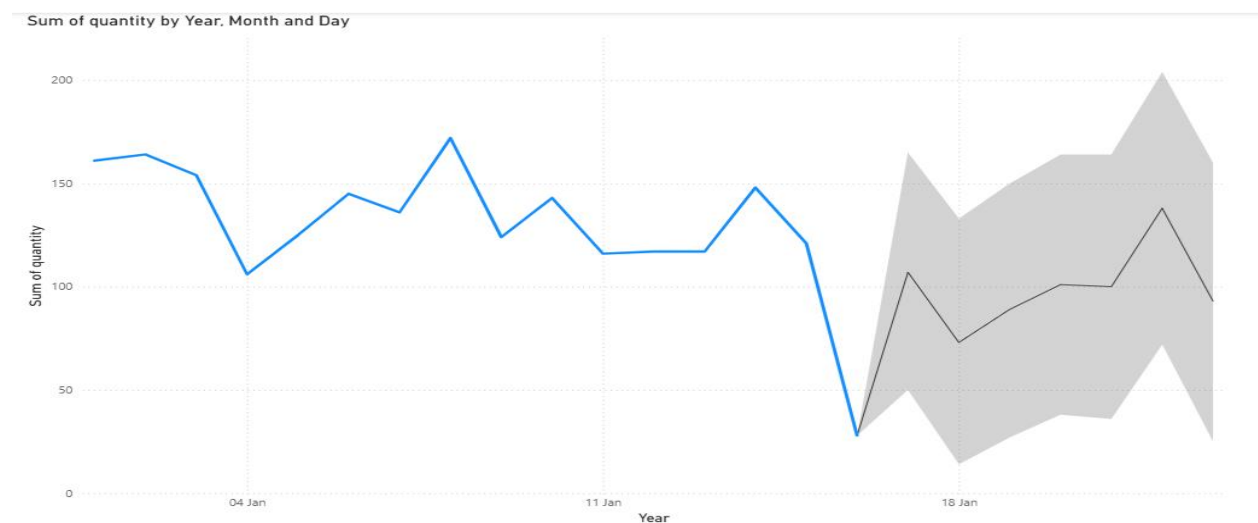


Fig.1

The above figure (**Fig.1**) depicts the demand of pizzas for the next 7 days. The shaded area represents the 7-day forecast for pizza demand. It shows an upward trend in pizza demand for the future week followed by a slight dip in pizza demand over the weekend.

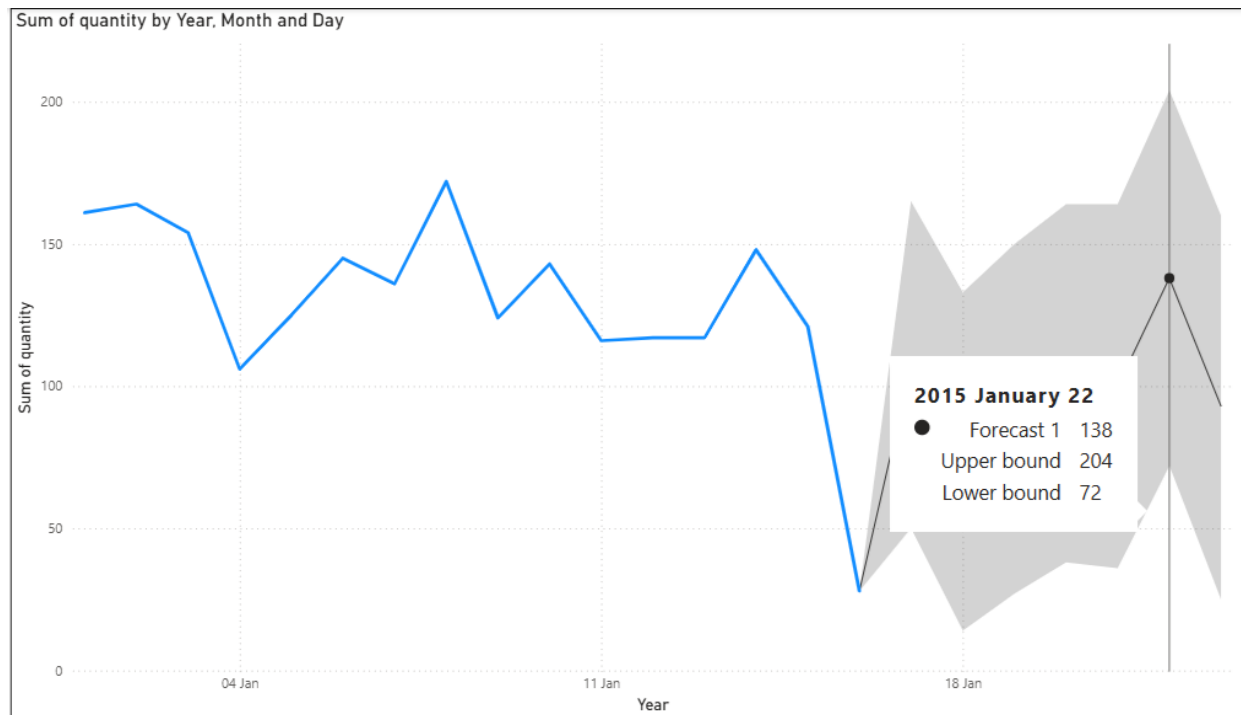


Fig.2

From the **Fig.2** above, we realize that the pizza demand has peaked on Thursday, January 22, 2015. The forecast shows the demand for pizzas as 138 on Thursday and it is the highest for the week. The upper bound for pizza demand is 204 and the lower bound for pizza demand is 72.

Based on the analysis, I have realized the following about the Pizza Sales Business:

- ❖ The Thai Chicken Pizza belonging to the Chicken category brings in more money compared to other categories. The Thai Chicken Pizza earns 1808.75\$ as Total Revenue.
- ❖ The Brie Carre Pizza is the most unpopular. It belongs to the Supreme category.
- ❖ On Thursdays, the Pizzeria is the busiest and at around 12 noon and 6 PM in the evening, the orders for pizzas peak.
- ❖ Pizzas of large size are very popular, and the most popular pizzas are five_cheese_l and thai_ckn_l.

9. Final Recommendations & Conclusion

Based on the analysis results, I have come up with the key business decisions regarding the following:

- **Staffing Optimization:** Increase kitchen staff on Thursdays at 12 noon and at 6 PM in the evening to handle peak order volumes.
- **Inventory Management:** Since Large pizzas drive the majority of sales, prioritize large-box and large-dough inventory.
- **Menu Engineering:** Consider bundling unpopular, low-volume pizzas with top-sellers via a promotional discount to clear slower-moving pizzas.

This Data Cleaning and Exploratory Data Analysis project successfully transformed raw pizza sales data into actionable business strategies, identifying Thursdays and Fridays as peak rush days of the week and Large pizzas as our primary growth drivers. By optimizing kitchen staffing during peak days and hours and by adjusting inventory for high-demand sizes, the restaurant can maximize efficiency and cut operational waste. The next step for this project is to connect this MySQL database to a visualization tool like Power BI to build a live performance dashboard.

